

Caleb Bellsmith

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TECHNICAL SKILLS

Languages: C++, HTML, LaTeX, Visual Basic, MATLAB

Software/Tools: SolidWorks, Fusion360, AutoCAD, VSCode, KiCad, CLion, Bamboo, Arduino IDE, GitHub, GitLab

Fabrication: CNC Machining, 3D Printing, Soldering, Manual Prototyping, Woodworking

EXPERIENCE

True North Bio-competition

Electrical Designer

2025 – Present

- Designing the electrical control framework for a lower-limb exoskeleton, using EMG sensors to facilitate intuitive, signal-based movement for assistive applications.

UWaterloo Space Soldering Team

Mechanical Designer

2025 – Present

- Designing mechanical housings and precision mounting structures for microgravity soldering mechanisms.

Biomechatronics Design Team

Mechanical Designer

2025 – Present

- Developing mechanical components for assistive orthotic devices, focusing on structural integrity and ergonomics.

BioTEC Conference

Sponsorship Lead

2025 – Present

- Securing partnerships and corporate outreach for Canada's largest student-run biotechnology conference.

PROJECTS

Hand Rehabilitation Exoskeleton

SolidWorks, ESP32, EMG Sensors

Independent Project

2025 – Present

- Created a mechanical orthotic exoskeleton aimed at rehabilitating hand injuries; optimized for minimal weight and tool-less adjustability. Developing motorized EMG based control for automation.

Auto Solving Maze

ESP32, C++, Computer Vision

Independent Project

2025 – Present

- Developed a servo-powered tilting platform using an ESP32 microcontroller. Implementing a webcam-based computer vision system to enable autonomous maze-solving and real-time pathfinding.

Custom Cedar Strip Canoe

CAD, CNC Machining

Personal Project

2024 – Present

- Computer-modeled and fabricated a custom canoe; integrated CNC-machined internal molds with traditional woodworking to achieve high structural efficiency and aesthetic finish.

Adjustable Finger Prosthetic

User Testing, CAD, Iterative Design

Independent Project

2025 – Present

- Developed a functional finger prosthetic for a patient with scleroderma through iterative user-testing and CAD.

EDUCATION & LEADERSHIP

University of Waterloo

Waterloo, ON

B.A.Sc. in Biomedical Engineering

- **WEEF Representative:** Nominated by class to evaluate technical proposals for student project funding and represented class interests.

Lifeguard & Swimming Instructor

Town of Milton

Instructor, NLS, CPR Certified

2023 – 2025

- Selected for competitive outdoor pool placement (Top 4% of 250+ staff) based on performance and leadership.

Distinctions and Awards

HHCS, TDCH

- Governor General's Academic Medal, Sir Isaac Newton Physics Award, TDCH Valedictorian, HHCS Valedictorian.